DEPARTMENT OF ELECTRICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY (ISM), DHANBAD

(MID SEMESTER EXAMINATION)

Exam: I-B.Tech. (Common) (A, B, C and D) **Session:** 2018-19

Sub: Electrical Technology (EEC 11101) **Semester:** Monsoon

Full Marks: 60 (Sixty) **Time:** 2 hrs

Instructions:

1) Diagrams should have proper labeling and variables should be defined.
2) Symbols and terms are carrying their usual meaning.

(Answer all questions)

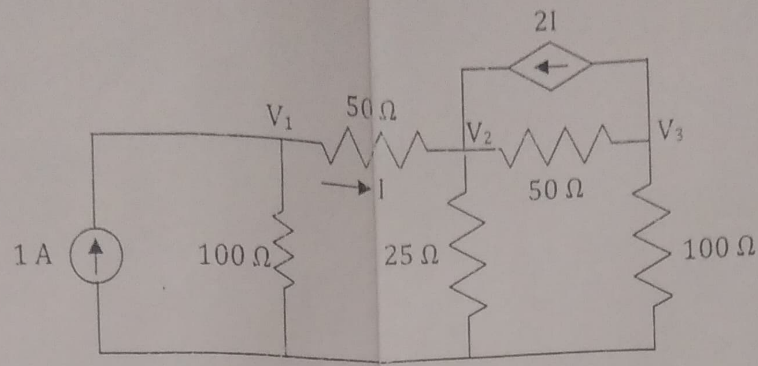
1. A resistance of 5Ω and an inductance of 50mH are connected in series to a 230V, 50Hz, single-phase a.c. supply. What value of capacitance is to be connected in series in this circuit to produce resonance at the supply frequency? Also, calculate the current taken from the supply at resonance. What is the quality factor of this R-L-C circuit? [4+3+3]

2. A resistor and an inductive coil in series are connected to 230V, 50Hz, single-phase a.c. mains. It is drawing a current of 5A at a lagging phase angle of 45^0 with respect to supply voltage. The magnitude of voltage across the coil is 163.25V. Determine (i) the resistance of the resistor: the resistance and inductance of the coil, (ii) the active power, reactive power and apparent power of the circuit, (iii) draw the phasor diagram of the circuit. [4+3+3]

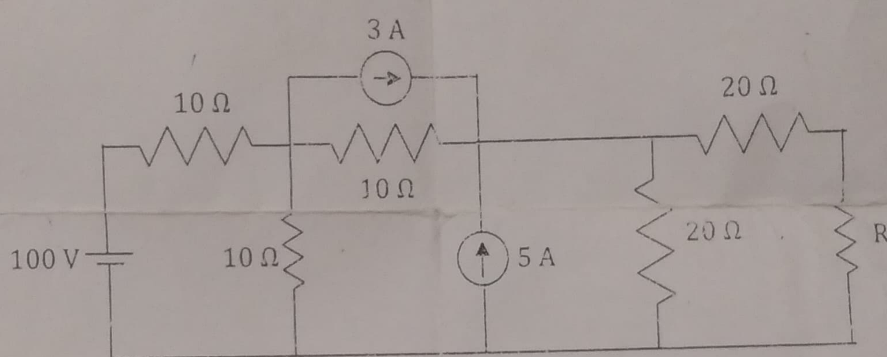
3. With suitable circuit diagram prove that the power consumed by a three-phase balanced load and its power factor can be measured by using two-wattmeter method. Also, draw the phasor diagram. [7+3]

[P.T.O.]

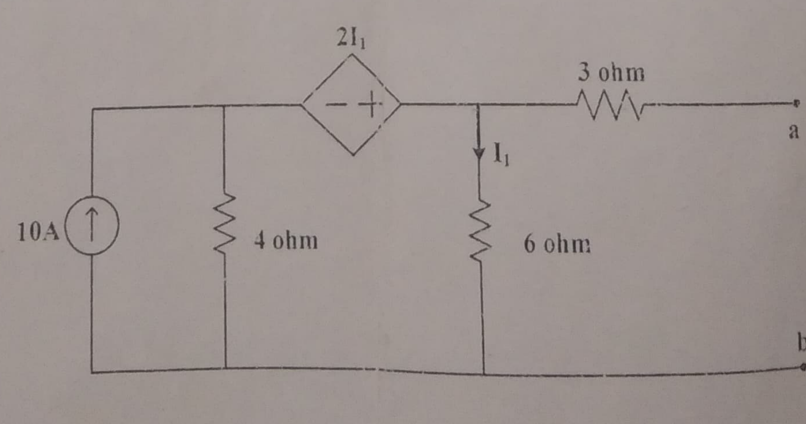
4. Determine the voltages V_1, V_2 and V_3 indicated in figure using Nodal Analysis. Find the power delivered or absorbed by the sources. [6+4]



5. Find the Norton's equivalent as viewed from the terminals of resistance R. [10]



6. Find the load resistance for which maximum power will be transferred to the load connected across the terminals a-b as shown in the figure. [10]



Theory